

Electronic Supplementary Material (ESM_1)

Can Arm-Mounted Piezoelectric Harvesting Extend Quadrotor Endurance? A Physics-Consistent Finite-Element and Energy-Aware Deep Reinforcement Learning Assessment

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This document contains supplementary tables and figures. All values are generated by scripts from the same simulation results used in the main article. The accompanying files are:

- **ESM_2.zip**: complete source code, trained policies and raw results (also at <https://github.com/omar123-1-cyber/UAV-Energy-Research>);
- **ESM_3_per_episode_results.csv**: outcome, energy, harvest, time and rotor speed for each of the 5,200 test episodes (25 learned policies $\times$ 200 scenarios, plus the A*+PD reference);
- **ESM_4_per_seed_summary.csv**: per-seed summary of every trained configuration.

S1 Finite-element verification

Table S1. Verification of the FE model (80 elements) against closed-form results.

Quantity	Reference	FE	Relative error (%)
f_1 , no tip mass	57.6768	57.6768	8.5e-6
f_2 , no tip mass	361.4544	361.4544	7.4e-6
Static tip deflection, 1 N (mm)	0.920053	0.920053	2.1e-7
f_1 with tip mass (Rayleigh)	18.4318	18.4291	0.01

Table S2. Mesh convergence of the first three bending frequencies (Hz) with the 73 g tip mass.

Elements	f_1	f_2	f_3
10	18.4291	259.7387	827.1841
20	18.4291	259.7347	827.0515
40	18.4291	259.7344	827.0431
80	18.4292	259.7344	827.0426
245	18.4262	259.7346	827.0425

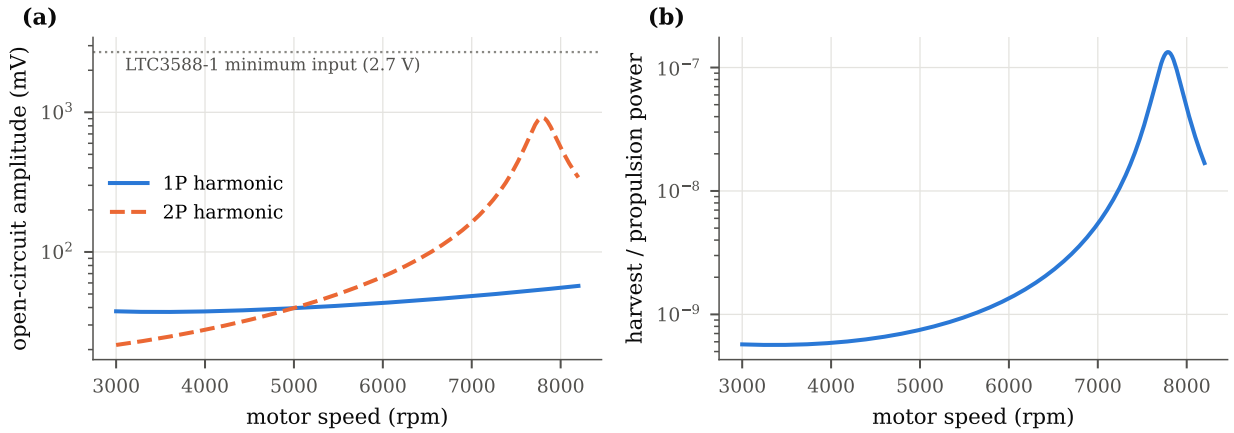


Figure S1. (a) Open-circuit voltage amplitude of the root patch for the 1P and 2P harmonics versus motor speed, compared with the 2.7 V minimum input of the LTC3588-1. (b) Ratio of four-patch harvest power to total propulsion power.

S2 Navigation results per seed

Table S3. Test results for every trained policy (200 held-out scenarios each). Energy, time, speed, rotor speed and harvest power are averaged over successful episodes.

Configuration	Seed	Success (%)	Crash (%)	Time-out (%)	Energy (J/m)	Time (s)	Speed (m/s)	Rotor speed (rpm)	Harvest (μ W)
SAC, energy-aware	0	82.5	17.0	0.5	87.7	5.14	1.91	5,085	0.119
	1	83.0	17.0	0.0	83.0	4.64	2.17	5,177	0.130
	2	78.5	21.5	0.0	84.4	4.90	2.12	5,085	0.120
	3	88.0	12.0	0.0	89.6	4.94	2.12	5,200	0.132
	4	86.0	13.5	0.5	83.9	4.85	2.14	5,118	0.122
PPO, energy-aware	0	62.0	38.0	0.0	64.9	3.84	2.59	5,051	0.120
	1	72.0	28.0	0.0	75.6	4.11	2.35	5,220	0.135
	2	77.5	22.5	0.0	77.3	4.20	2.46	5,214	0.134
	3	76.5	23.0	0.5	73.7	4.12	2.37	5,173	0.128
	4	74.0	26.0	0.0	70.7	3.85	2.50	5,222	0.134
DQN, energy-aware	0	67.5	32.5	0.0	106.3	5.77	1.83	5,190	0.139
	1	56.5	40.0	3.5	99.2	5.54	1.83	5,183	0.138
	2	71.0	28.5	0.5	118.2	6.72	1.73	5,159	0.137
	3	37.5	39.0	23.5	122.2	6.51	1.67	5,164	0.137
	4	65.0	29.0	6.0	118.8	6.63	1.73	5,154	0.136
SAC, standard	0	83.5	16.0	0.5	113.0	6.59	1.72	5,114	0.119
	1	72.5	23.0	4.5	98.1	5.59	1.91	5,163	0.125
	2	83.0	15.0	2.0	96.9	5.56	1.97	5,133	0.124
	3	79.0	17.5	3.5	101.7	5.71	1.86	5,169	0.126
	4	89.0	11.0	0.0	84.3	4.99	2.07	5,081	0.118
SAC, energy-aware + harvest bonus	0	89.5	10.5	0.0	96.1	5.67	1.87	5,079	0.118
	1	85.5	14.5	0.0	99.2	5.79	1.85	5,102	0.120
	2	84.5	15.5	0.0	87.7	5.18	2.01	5,081	0.119
	3	83.0	17.0	0.0	104.5	5.97	1.83	5,145	0.125
	4	79.0	21.0	0.0	95.1	5.66	1.82	5,074	0.117
A*+PD (exact map)	–	100.0	0.0	0.0	95.3	5.53	1.57	5,122	0.118

Table S4. Robustness of the five energy-aware SAC policies (success in %, first 100 test scenarios, no retraining).

Condition	Seed 0	Seed 1	Seed 2	Seed 3	Seed 4	Mean	A*+PD
nominal	81	84	76	83	90	82.8	100.0
ray noise $\sigma=0.05$	76	85	72	83	89	81.0	100.0
ray noise $\sigma=0.10$	79	85	72	86	88	82.0	100.0
wind $\sigma=0.5$	73	75	73	78	75	74.8	76.0
wind $\sigma=1.0$	59	54	48	60	53	54.8	48.0
noise 0.05 + wind 0.5	77	79	71	80	74	76.2	75.0



Figure S2. Validation success during training for each seed and configuration (30 validation scenarios every 10^4 steps).

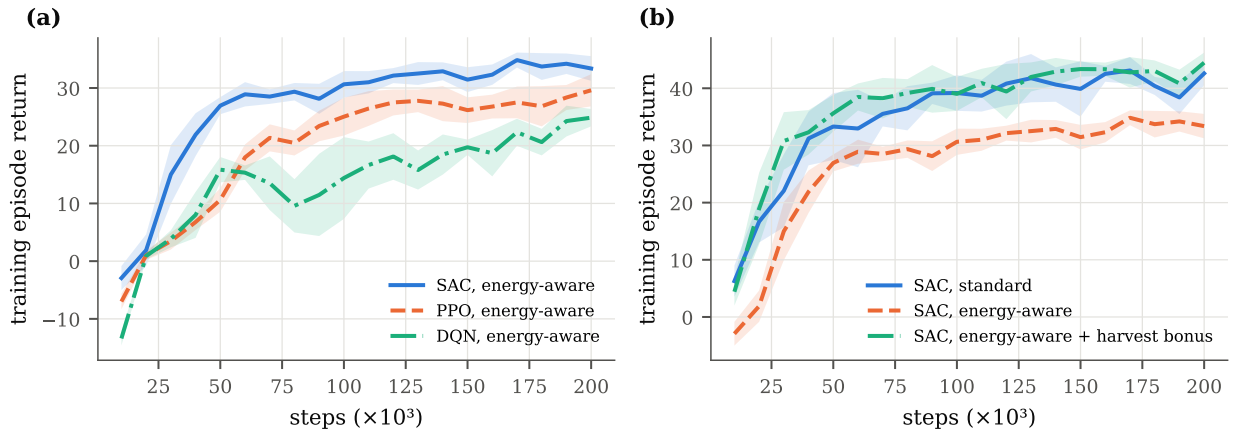


Figure S3. Training episode return (mean of the last 50 episodes; mean $\pm$ s.d. over five seeds). (a) Algorithms with the energy-aware reward. (b) SAC with the three reward configurations. Returns of different reward configurations are not directly comparable, because the reward terms differ.

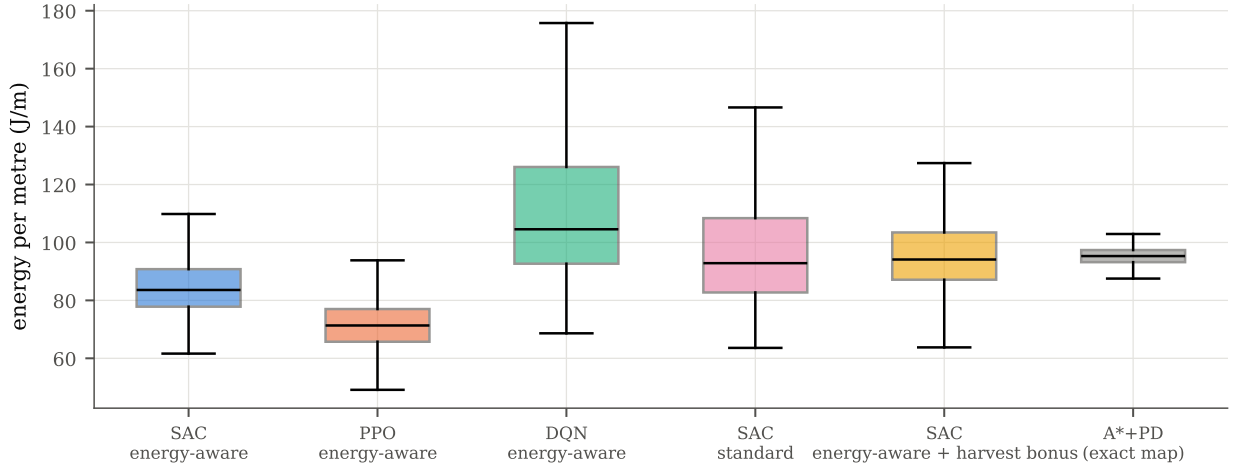


Figure S4. Distribution of propulsion energy per metre of goal distance over all successful test episodes (five seeds pooled; boxes show quartiles, whiskers 1.5 IQR, outliers omitted).

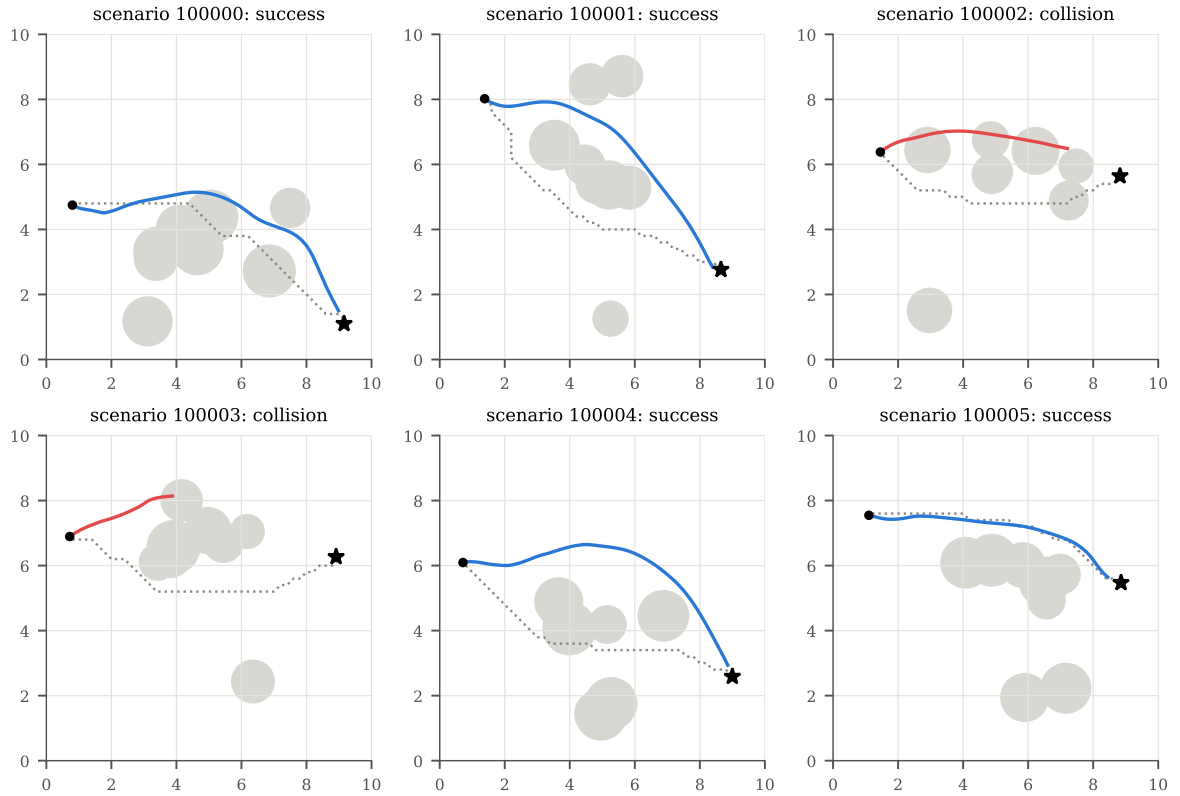


Figure S5. Top view of the first six test scenarios with the energy-aware SAC policy (seed 0), shown unselected, including failures. Blue: successful flight; red: failed flight; dotted grey: A* path computed with the exact map. Obstacles are drawn as horizontal projections.